

ORIGINAL ARTICLE

Timing of Orthostatic Blood Pressure Changes and Its Association With Orthostatic Symptoms and Falls at Very Old Ages: Findings From the ARIC Study

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BACKGROUND: Assessment and treatment of orthostatic hypotension (OH) are recommended to prevent falls in older adults. However, the optimal timing of standing blood pressure (BP) assessments to identify clinically relevant OH remains unclear.

METHODS: We measured 3 supine BP readings after 5 minutes of rest at 30-second intervals and 6 standing BP readings (timed at 0, 1, 2, 3, 4, and 5 minutes after standing) during the ARIC study (Atherosclerosis Risk in Communities) visit 10. OH was defined as a drop in BP (systolic ≥ 20 mmHg or diastolic ≥ 10 mmHg) on standing from the supine position. We quantified the prevalence and predictors of initial OH (immediately after standing), early OH (within 3 minutes), late OH (after 3 minutes), and sustained OH (all 6 minutes). We examined their associations with orthostatic symptoms and falls using logistic regression and negative binomial models, respectively.

RESULTS: Of the 863 participants (mean age, 83.7 years), 41.3% had initial OH. Participants on antihypertensive medications had higher odds of all types of OH irrespective of controlled ($<130/80$ mmHg) or uncontrolled ($\geq 130/80$ mmHg) hypertension compared with those without hypertension ($<130/80$ mmHg) and no antihypertensive use. Initial and early OH were strongly associated with orthostatic symptoms in the process of standing from the supine position (odds ratio, 2.34 [95% CI, 1.42–3.86]; odds ratio, 1.82 [95% CI, 1.11–2.98], respectively). Finally, participants with initial OH had higher fall rates (rate ratio, 1.45 [95% CI, 1.02–2.06]) than those without OH.

CONCLUSIONS: Measuring BP within 3 minutes of standing, particularly the first BP measurement immediately after standing, may provide clinicians with important information on falls risk and orthostatic symptoms. (*Hypertension*. 2026;83:00–00.

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Key Words: aged ■ antihypertensive agents ■ hypertension ■ hypotension, orthostatic ■ supine position

Orthostatic hypotension (OH), defined as a drop in blood pressure (BP) on standing, is a common observation among older populations, particularly those with hypertension.^{1–3} OH is estimated to affect nearly one-quarter of adults aged ≥ 60 years.³ OH is strongly associated with falls, fractures, dementia, and

adverse cardiovascular events, such as heart attack, stroke, and death.^{2,4–8}

Traditionally, OH is diagnosed by taking multiple BP measurements in the supine, seated, and standing positions over several minutes.^{5,9} Although general guidelines recommend measuring supine or seated BP after

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NOVELTY AND RELEVANCE

What Is New?

In this community-based cohort of older adults (mean age, 83.7 years), orthostatic hypotension within 3 minutes of standing, especially immediately after standing, was common and strongly associated with antihypertensive medications, orthostatic symptoms, and falls.

What Is Relevant?

Identifying the timing of standing blood pressure measurement is critical for assessing clinically relevant orthostatic hypotension and improving hypertension management among older adults.

Clinical/Pathophysiological Implications?

Clinicians should consider measuring standing blood pressures at least once immediately after standing to identify orthostatic hypotension in older adults as this may provide important information on orthostatic symptoms and falls.

Nonstandard Abbreviations and Acronyms

ARIC	Atherosclerosis Risk in Communities
BMI	body mass index
BP	blood pressure
OH	orthostatic hypotension
SBP	systolic blood pressure
SPRINT	Systolic Blood Pressure Intervention Trial
STURDY	Study to Understand Fall Reduction and Vitamin D in You

5 minutes of rest,^{10,11} there is substantial variation in recommendations regarding when standing measurements should be performed.⁵ Several guidelines recommend using the average of BP measured within 3 minutes of standing to diagnose OH.^{4,5,9} However, in recent work among a population of older adults aged ≥ 70 years, OH detected within 1 minute of standing was strongly associated with orthostatic symptoms, such as lightheadedness and dizziness,^{8,12} as well as with falls in older adults with uncontrolled hypertension.¹ Late OH, occurring after 3 minutes of standing, has also been reported to be strongly associated with falls.¹² However, these findings have not been replicated in very old populations. Moreover, risk factors for early versus late OH remain unclear.

In the present study, we used timed BP measurements in the standing position to assess OH among participants of the 10th study visit of the ARIC cohort study (Atherosclerosis Risk in Communities). The aims of this study were to (1) characterize changes in BP when transitioning from the supine to standing position; (2) investigate the prevalence and predictors of different OH phenotypes in very old ages, such as OH immediately after standing, within 3 minutes, after 3 minutes, and during the entire 6 minutes of standing; (3) identify the associations of different OH phenotypes with orthostatic

symptoms, such as lightheadedness, dizziness, fainting or blackout, and imbalance, experienced in the last 30 days or during the OH assessment; and (4) examine how OH phenotypes were associated with falls in the last 12 months and during follow-up after completing the OH assessment.



METHODS

Data Availability

Because of the sensitive nature of the data collected for this study, requests to access the data set from qualified researchers trained in human subject confidentiality protocols may be sent to the ARIC Coordinating Center or Biologic Specimen and Data Repository Information Coordinating Center (BioLINCC).¹³

Study Design and Population

The ARIC study is a prospective cohort study of 15792 adults aged 45 to 64 years who were recruited in 1987 from 4 communities in the United States: Forsyth County, NC; Jackson, MS; suburban Minneapolis, MN; and Washington County, MD.^{14,15} To date, 11 follow-up visits have been conducted among ARIC participants, and the 12th visit is ongoing. All participants of visit 10 (mean age, 83.7 years) were invited to participate in an ancillary study to investigate OH in 2023. Details on visit 10 participant recruitment and safety screening were provided in the ARIC Study Manual of Operations (see [Supplemental Methods](#) for participants' eligibility criteria).¹⁶ A total of 1044 participants completed OH assessments. The present analysis was further restricted to 863 participants after excluding those with missing relevant covariates ($n=143$), missing OH phenotypes ($n=19$), and missing information to assess orthostatic symptoms ($n=19$).

Standard Protocol Approvals, Registrations, and Patient Consents

Written informed consent was obtained from all participants, and the study protocol was approved by a single institutional review board at Johns Hopkins University, used by all study

sites. The Beth Israel Deaconess Medical Center Institutional Review Board considered this analysis of deidentified data to be exempt from human subject research oversight.

Orthostatic Hypotension

Participants underwent 3 supine BP measurements, each separated by a 30-second interval, after an initial 5 minutes of rest, using an automated oscillometric device (Omron HEM-907XL).¹⁷ This device was used in the SPRINT study (Systolic Blood Pressure Intervention Trial) and studied elsewhere for OH among older adults.¹⁸ We calibrated the device every 6 months using the SimCube SC-4Kit noninvasive BP simulator¹⁹ following the study protocol guidelines. Then, 6 stopwatch-timed BP measurements were taken during 6 minutes after standing from the supine position at 1-minute intervals using the same device (see [Supplemental Methods](#) for details on the OH assessment). We calculated the difference between standing BPs and the average of supine BPs. OH was determined if either a drop in systolic BP (SBP) ≥ 20 mmHg or diastolic BP ≥ 10 mmHg occurred after standing from the supine position. We defined the following OH phenotypes based on prior work^{4,5,8,9,12} ([Figure S1](#)): (1) initial: OH during the first minute immediately after standing; (2) early: OH during the first 3 minutes of standing (based on the average of the first 3 standing BPs); (3) late: OH after 3 minutes of standing (based on the average of the last 3 standing BPs); (4) sustained: OH during the entire 6 minutes of standing (if both early and late OH were present). These OH phenotypes were not mutually exclusive ([Table S1](#)).

Orthostatic Symptoms and Falls

In this study, orthostatic symptoms included the following: lightheadedness, dizziness, fainting, blackout, and imbalance. Participants were asked how often they experienced each of these symptoms while standing up in the past 30 days (historic). Responses were recorded using a 5-point Likert scale (1: never, 2: rarely, 3: sometimes, 4: often, and 5: every time). For analysis, we dichotomized the responses as no (1) and yes (2–5), with a sensitivity analysis using no (1–2) and yes (3–5). Concurrent orthostatic symptoms were based on participant responses regarding whether they experienced any orthostatic symptoms in the process of standing or at any time during the 6 minutes while in the standing position. Responses were similarly recorded on a 5-point Likert scale and dichotomized as no (1) and yes (2–5). Details are provided in [Supplemental Methods](#).

Because OH may represent a chronic and recurrent condition,²⁰ we examined the associations of OH phenotypes with falls in the last 12 months (historic) and falls during follow-up after the OH assessment (prospective). Historic falls were based on participant responses in the OH assessment to the question of whether they had experienced a fall in the last 12 months. Participants were asked to provide the approximate date of these falls. Prospective falls were obtained from the ARIC annual telephone follow-up conducted after visit 10. They were asked, "In the past 12 months, did you fall? (yes or no)." A fall was defined as a sudden, unintentional change in position that causes landing at a lower level, on an object, the floor, or the ground.^{21–25} Participants were also instructed not to include falls that occurred due to sudden paralysis, a seizure, a stroke, syncope or fainting, or being pushed. If they endorsed falling, they were asked, "In the past 12 months, how many

times did you fall?" In cases where visit 10 occurred within the 12 months referred to in the annual follow-up question and the duration of historic falls partly overlaps with that of prospective falls, we used the historic fall date to distinguish between historic and prospective falls. However, if there was a discordance between historic and prospective falls during the overlapped duration (ie, participants reported historic but no prospective falls), we assumed no prospective falls since visit 10. For each participant, follow-up duration was calculated as the interval between the OH assessment completion date and the post-OH assessment follow-up date.

Covariates

Data on sociodemographic characteristics were collected at ARIC visit 1, whereas comorbidities, activities, and medication use were collected at each ARIC study visit by trained personnel using standardized protocols maintaining extensive quality control measures. Covariates of interest were selected a priori based on literature review and biological knowledge for hypothesized relationships with OH and orthostatic symptoms. Covariates included age (in years), body mass index (BMI; kg/m²), time required to stand from the supine position (in seconds), biological sex (male or female), diabetes (yes or no), frailty (frail, prefrail, or robust²⁶), prior coronary heart disease (yes or no), prior stroke (yes or no), and hypnotic use (yes or no). Information on race, self-reported Black or White race, was combined with study center to create a race-center variable as follows: Washington County–White, Jackson–Black, Minneapolis–White, Forsyth–Black, and Forsyth–White. BMI was calculated using height (in meters) and weight (in kilograms) measurements. Hypertension (ie, stage 1 or undiagnosed) was defined as SBP ≥ 130 mmHg or diastolic BP ≥ 80 mmHg, based on the mean of 3 seated BP measurements measured at 60-second intervals after 5 minutes of seated rest using an Omron HEM-907XL device. Participants were asked to bring all medications taken during the previous 4 weeks to the clinic visit. Antihypertensive and hypnotic use were based on Medi-Span Generic Product Identifier codes. Hypertension and antihypertensive medication use were used to create 4 categories: (1) uncontrolled hypertension ($\geq 130/80$ mmHg) and antihypertensive use, (2) hypertension ($\geq 130/80$ mmHg) but no antihypertensive use, (3) controlled hypertension ($< 130/80$ mmHg) and antihypertensive use, and (4) no hypertension ($< 130/80$ mmHg) and no antihypertensive use.

Frailty was defined as previously reported in the ARIC study²⁶: (1) weight loss, defined as a 10% weight loss from visit 9 to visit 10 or a visit 10 BMI < 18.5 kg/m²; (2) exhaustion was defined using the self-reported responses to the Center for Epidemiological Studies Depression scale. Participants were considered exhausted if they responded with "some of the time" or "most of the time" to either of the following statements: "I felt everything I did was an effort" or "I could not get going"; (3) slowness, classified as walking speed in the bottom 20th percentile for sex and height, measured during a 4-m walk test; (4) low grip strength, defined as the lowest 20th percentile of grip strength, adjusted for sex and BMI; and (5) low energy expenditure, defined as the lowest 20th percentile of the sex-specific leisure sports activity index, measured by the Baecke physical activity questionnaire.²⁷ In ARIC visit 10, leisure sports activity index was calculated differently from previous ARIC visits due to the lack of certain physical activity data (see [Supplemental Methods](#)).

Finally, an individual was defined as frail if ≥ 3 components were present, prefrail if 1 or 2 components were present, and robust if no components were present.

Statistical Analysis

We presented the baseline characteristics of all participants using arithmetic means and proportions for continuous and categorical variables, respectively. We estimated the mean of 3 supine SBPs and the mean of each of the 6 standing SBPs, using mixed model for repeated measurements, and stratified by antihypertensive medication use and OH phenotypes. We investigated changes in SBP from the supine to standing position by plotting these means graphically. In addition, we examined mean differences in supine SBP between participants with and without OH using mixed model for repeated measurements. These analyses were repeated for diastolic BP.

We used logistic regression to identify predictors of OH phenotypes and their associations with historic and concurrent symptoms. Negative binomial models accounting for overdispersion were used to examine associations of each OH phenotype with historic falls, using 1 year as the offset, and with prospective falls, using follow-up time as the offset. All models were adjusted for the listed covariates.

In supplemental analyses, we used individual measurements to redefine (1) initial OH if it occurred during the first minute immediately after standing and then resolved during the second through sixth minutes, and (2) sustained OH if OH was present at every individual measurement during the entire 6 minutes of standing position (Figure S1). In addition, we performed the following sensitivity analyses: (1) considering participants having OH immediately after standing if they missed the first standing measurement but reported orthostatic symptoms at any time during the 6 minutes of standing position⁷; (2) using falls reported in the last 6 months; and (3) excluding participants who had to sit immediately after standing due to symptoms or risk of falls ($n=18$). All analyses were performed using SAS version 9.4.²⁸ Results were considered statistically significant if the 95% CI did not include the null value.

RESULTS

Participant Characteristics

Among 863 participants (mean age, 83.7 years, BMI, 27.0 kg/m²), 59.2% were female, 31.8% had diabetes, and 4.4% were frail. Participants required an average of 9.4 seconds to stand up from the supine position. Approximately 90% of participants had either uncontrolled ($\geq 130/80$ mmHg) or controlled ($<130/80$ mmHg) hypertension, and 80.5% used antihypertensive medications (Table 1). Participants' characteristics were similar across OH phenotypes, orthostatic symptoms, and falls (Tables S2 through S4).

Prevalence of OH, Orthostatic Symptoms, and Falls

Initial OH was present in 41.3% of study participants ($N=863$), representing the most prevalent type of OH

in our study. One-fourth of the participants (23.4%) had early OH, whereas late and sustained OH were 18.0% and 15.1%, respectively (Table 1). Around 10% of participants reported experiencing any orthostatic symptoms in the process of standing, whereas 5.7% reported experiencing symptoms at any time during the 6 minutes of the standing position. Imbalance was the most common reported symptom (62.3%) experienced in the last 30 days. Participants had a total of 303 falls during the last 12 months before the OH assessment (historic), with 25.8% of participants having at least 1 fall, whereas 296 falls (prospective) occurred during follow-up after the OH assessment (mean duration of follow-up, 1 year), with 22.1% of participants having at least 1 fall (Table 1). Overall, participants with any OH had a higher number of historic or prospective falls than those without OH (Table S2).

Changes in BP From Supine to Standing

Study participants had an average supine BP of 140/70 mmHg, with a mean drop of 15/7 mmHg immediately after standing. However, average BP recovered within minutes and stabilized (Figure [A]; Figure S2A). Those on antihypertensive medications had slightly lower supine and standing BP than nonantihypertensive users (Figure [B]; Figure S2B).

In the case of all OH types, participants with OH had significantly higher supine SBP or diastolic BP compared with those without OH (Figure [C] through [F]; Figure S3; Tables S5 and S6). However, supine BP varied across OH phenotypes. Individuals with sustained OH had a higher average supine BP (155/74 mmHg) than those with initial OH (145/72 mmHg). On standing, sustained OH had the largest drop in BP, whereas initial OH had the smallest (Figure [C] and [F]; Figure S3A and S3D).

Predictors of Different OH Phenotypes

Participants on antihypertensive medications were more likely to have initial and early OH irrespective of uncontrolled ($\geq 130/80$ mmHg) or controlled ($<130/80$ mmHg) hypertension. Those with controlled hypertension on antihypertensive medications had higher odds of late and sustained OH than nonhypertensive ($<130/80$ mmHg) and nonmedication users (Table 2). Male sex was significantly associated with initial OH compared with female sex. No clear patterns emerged between any of the other factors considered and OH phenotypes.

Associations of OH With Orthostatic Symptoms

Participants with initial OH had significantly higher odds of experiencing any orthostatic symptoms during the

Table 1. Characteristics of Study Participants (N=863)

Characteristics	N (%)
Age,* y	83.7 (3.7)
BMI,* kg/m ²	27.0 (4.8)
Time to stand from the supine position,* s	9.4 (9.1)
Female	511 (59.2)
Race-study center	
Washington County (White)	243 (28.2)
Jackson (Black)	130 (15.1)
Minneapolis (White)	276 (32.0)
Forsyth (Black)	18 (2.1)
Forsyth (White)	196 (22.7)
Diabetes	274 (31.8)
Frailty	
Frail	38 (4.4)
Prefrail	547 (63.4)
Robust	278 (32.2)
Prior coronary heart disease	96 (11.1)
Prior stroke	27 (3.1)
Hypertension and antihypertensive use	
Uncontrolled hypertension and antihypertensive use	367 (42.5)
Hypertension but no antihypertensive use	81 (9.4)
Controlled hypertension and antihypertensive use	328 (38.0)
No hypertension and no antihypertensive use	87 (10.1)
Hypnotic use	81 (9.4)
Initial OH†	319 (41.3)
Early OH	206 (23.4)
Late OH	155 (18.0)
Sustained OH	130 (15.1)
Symptoms during the OH assessment (concurrent)	
Symptoms in the process of standing	87 (10.1)
Symptoms at any time during the 6 min of standing position	49 (5.7)
Symptoms in the last 30 d (historic)	
Lightheadedness	335 (38.8)
Dizziness	292 (33.8)
Fainting or blackout	29 (3.4)
Imbalance	538 (62.3)
Falls in the last 12 mo (historic)	238 (27.6)
No. of falls in the last 12 mo (historic)†	
0 fall	625 (74.1)
1 fall	158 (18.7)
≥2 falls	60 (7.1)
Total no. of falls in the last 12 mo (historic)†	303
Falls during follow-up (prospective)†	
0 fall	601 (77.9)
1 fall	108 (14.0)
≥2 falls	63 (8.2)
Total number of falls during follow-up (prospective)†	296

(Continued)

Table 1. Continued

Characteristics	N (%)
Post-OH assessment follow-up,*† y	1.0 (0.4)

OH: SBP drops ≥ 20 mm Hg or DBP ≥ 10 mm Hg on standing from the supine position. Initial OH: OH occurred during the first minute immediately after standing. Early OH: OH occurred during the first 3 min of standing, based on the average of the first 3 BP measurements. Late OH: OH occurred during the last 3 min of standing, based on the average of the last 3 BP measurements. Sustained OH: OH present during the entire 6 min of standing, based on the averages of the first and last 3 measurements (ie, both early and late OH were present). Hypertension: SBP ≥ 130 mm Hg or DBP ≥ 80 mm Hg. Uncontrolled hypertension: Hypertension and antihypertensive medication use. Controlled hypertension: No hypertension and antihypertensive medication use. No hypertension: SBP < 130 mm Hg and DBP < 80 mm Hg. Symptoms in the process of standing: based on responses to "Did you feel dizziness, lightheadedness, faint, or like you might blackout in the process of standing up?" Please rate on a scale from 1 to 5, with 1 being "no symptoms" and 5 being the "worst possible." (1=no symptoms, 2=some, 3=mild, 4=moderate, and 5=worst possible). Recategorized for the analysis: no=no symptoms and yes=some to worst possible. Symptoms at any time during the 6 min of standing position: based on responses to "Did you feel dizziness, lightheadedness, faint, or like you might blackout at any time while standing?" (1=no symptoms, 2=some, 3=mild, 4=moderate, 5=worst possible). Recategorized for the analysis: no=no symptoms and yes=some to worst possible. Symptoms in the last 30 d: based on responses to "We would like to ask you questions about symptoms you have experienced in the past 30 d during the process of standing up." On a scale from 1 to 5, where 1 represents "never" and 5 represents "every time you stand without exception," please rate how often you experience the following: lightheadedness, dizziness, fainting, blackout, imbalance (1=never, 2=rarely, 3=sometimes, 4=often, and 5=every time). Recategorized for the analysis: no=never and yes=rarely to every time. BMI indicates body mass index; BP, blood pressure; DBP, diastolic blood pressure; OH, orthostatic hypotension; and SBP, systolic blood pressure.

*Mean (SD)

†n for initial=773, n for number of falls in the last 12 mo=843, and n for falls during follow-up=772.

process of standing (odds ratio, 2.34 [95% CI, 1.42–3.86]); however, no significant association was observed with symptoms experienced at any time during the 6-minute standing period. In addition, they were more likely to experience lightheadedness (odds ratio, 1.48 [95% CI, 1.08–2.02]) in the last 30 days (Table 3). However, individuals with early OH tended to experience any orthostatic symptoms in the process of standing and at any time during 6 minutes in the standing position (odds ratio, 1.82 [95% CI, 1.11–2.98] and odds ratio, 1.94 [95% CI, 1.02–3.69], respectively).

Associations of OH With Falls

Participants with initial OH were likely to have falls (prospective) during follow-up (rate ratio, 1.45 [95% CI, 1.02–2.06]) compared with those without OH. However, no significant association was observed between any OH phenotypes and falls in the last 12 months (historic; Table 4). It is noteworthy that participants who missed the first BP measurement immediately after standing had significantly higher odds of historic falls (Table S7). However, our sensitivity analysis considering participants as having OH if they missed the first measurement and reported experiencing orthostatic symptoms during the OH assessment did not change the relationship between initial OH and historic falls (Table S8).

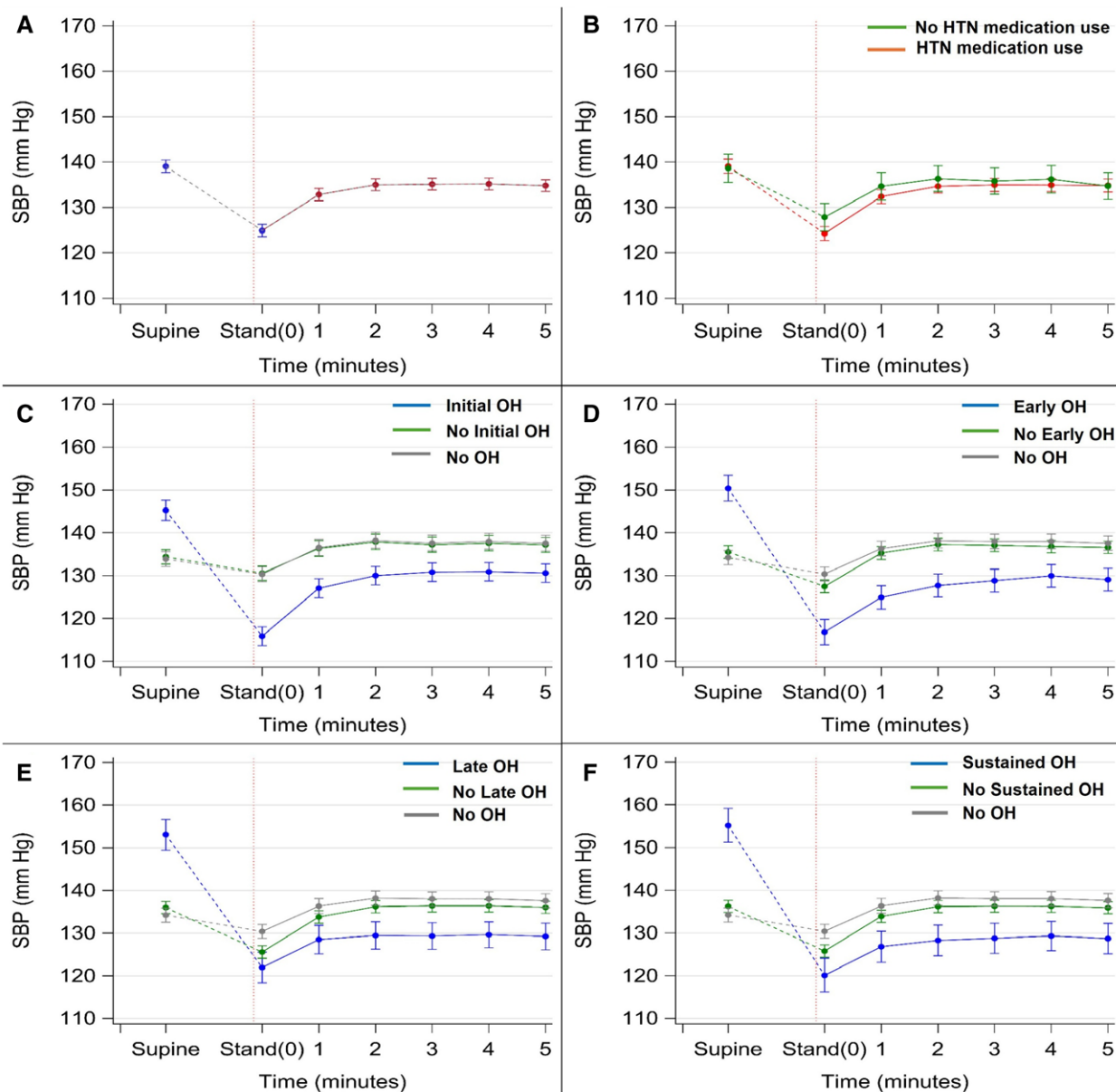


Figure. Xxx.

Patterns of mean systolic blood pressure (SBP) changes over time while transitioning from the supine to standing position among (A) all participants, (B) participants stratified by antihypertensive (hypertension [HTN]) medication use, and across orthostatic hypotension (OH) phenotypes, (C) initial OH, (D) early OH, (E) late OH, and (F) sustained OH. Supine SBP was based on 3 blood pressure measurements. The vertical dashed red line indicates the time point when participants began standing from the supine position. Plotted means and their 95% CIs were estimated by mixed model for repeated measurements (restricted maximum likelihood ratio, unstructured covariance).

Supplement and Sensitivity Analyses

Initial and sustained OH redefined using individual measurements had almost similar findings to initial and sustained OH defined using averages (Figure S4; Tables S9 through S11). Using falls reported in the last 6 months before the OH assessment (Table S12) or excluding the seated BP measurements (Tables S13 through S15) from the analytic sample did not meaningfully change study findings compared with the findings of the main analysis. When responses for historic symptoms were

recategorized, nearly all OH phenotypes were significantly associated with lightheadedness and dizziness, indicating no differential effect of the timing of orthostatic BP changes on historic symptoms (Table S16).

DISCUSSION

In this community-based population of older adults with an average age of 83.7 years, initial OH was the most prevalent (41.3%). Participants with any OH phenotype

Table 2. Predictors of Different Phenotypes of Orthostatic Hypotension (OH; N=863*)

Characteristics	Odds ratio (95% CI)			
	Initial OH*	Early OH	Late OH	Sustained OH
No. with type of OH, %	319 (41.3)	206 (23.4)	155 (18.0)	130 (15.1)
Age, y	1.03 (0.99–1.07)	1.02 (0.97–1.06)	1.02 (0.97–1.07)	1.02 (0.97–1.08)
BMI, kg/m ²	0.97 (0.94–1.01)	0.97 (0.94–1.01)	0.96 (0.92–1.00)	0.97 (0.93–1.01)
Time to stand from the supine position, s	1.01 (0.98–1.03)	1.00 (0.98–1.02)	1.01 (0.99–1.03)	1.01 (0.99–1.03)
Sex				
Male	1.38 (1.01–1.87)†	1.34 (0.96–1.88)	1.38 (0.95–2.01)	1.26 (0.85–1.88)
Female	Reference	Reference	Reference	Reference
Diabetes				
Yes	0.88 (0.63–1.23)	0.93 (0.65–1.34)	1.13 (0.75–1.69)	0.92 (0.60–1.42)
No	Reference	Reference	Reference	Reference
Frailty				
Frail	1.60 (0.74–3.46)	1.69 (0.77–3.71)	1.19 (0.47–2.99)	1.48 (0.59–3.71)
Prefrail	1.24 (0.90–1.73)	1.28 (0.89–1.85)	1.35 (0.90–2.04)	1.29 (0.83–2.01)
Robust	Reference	Reference	Reference	Reference
Prior coronary heart disease				
Yes	1.05 (0.64–1.72)	0.70 (0.41–1.20)	0.74 (0.40–1.35)	0.84 (0.45–1.57)
No	Reference	Reference	Reference	Reference
Prior stroke				
Yes	1.66 (0.69–3.96)	1.09 (0.44–2.69)	0.62 (0.20–1.89)	0.83 (0.27–2.53)
No	Reference	Reference	Reference	Reference
Hypertension and antihypertensive medication use				
Uncontrolled hypertension and antihypertensive use	1.89 (1.07–3.32)†	2.57 (1.29–5.12)†	1.64 (0.79–3.41)	1.84 (0.82–4.11)
Hypertension but no antihypertensive use	1.15 (0.56–2.39)	1.05 (0.42–2.65)	1.55 (0.63–3.82)	0.98 (0.33–2.88)
Controlled hypertension and antihypertensive use	1.98 (1.12–3.50)†	3.21 (1.60–6.43)†	2.47 (1.19–5.14)†	2.68 (1.20–5.98)†
No hypertension and no antihypertensive use	Reference	Reference	Reference	Reference

There are 4 models presented in this table. Each model included all characteristics as independent variables (predictors) and an OH phenotype as the dependent variable. All models were adjusted for race–study center and hypnotic use using logistic regression. Results were considered statistically significant if the 95% CI did not include the null value. BMI indicates body mass index.

*n=773 (90 participants missed the first standing measurements).

had higher supine BP than those without OH. Male participants were more likely to have initial OH than female participants. Antihypertensive medication use was strongly associated with initial and early OH, irrespective of uncontrolled ($\geq 130/80$ mm Hg) or controlled ($<130/80$ mm Hg) hypertension. These OH phenotypes were also associated with orthostatic symptoms during the OH assessment. Finally, participants with initial OH were more likely to have falls during follow-up. These findings reinforce the importance of monitoring for orthostatic changes soon after standing among older adults, especially among male adults and those treated for hypertension.

The prevalence of OH varies by age, clinical conditions,^{3,12,29–31} and the timing of standing BP measurements. Consistent with prior studies, we observed a higher prevalence of OH immediately after standing (41.3% initial OH) than after 3 minutes of standing (18.0% late OH).^{12,29} Although several studies have suggested that BP measurement immediately after standing should not

be used to define OH due to wide variability,^{1,18,32–36} our findings suggest that excluding these measurements would substantially underestimate OH and miss clinically relevant information from initial OH, which has a strong association with orthostatic symptoms and falls among the ARIC population.⁸

Orthostatic BP responses may differ by biological sex, which is consistent with our study findings that showed higher odds of initial OH among older male adults compared with female adults. This contrasts with younger populations, in which female adults tend to have greater orthostatic intolerance, possibly due to smaller stroke volumes, attenuated sympathetic response, and increased vasodilatory capacity.³⁷ However, prior studies reported a higher prevalence of OH among elderly male (aged >75 years) than female participants.³⁸ Although Schumann et al³⁹ reported a steeper decline in baroreflex sensitivity among females aged ≥ 60 years likely due to the loss of estrogen's protective effects, OH remains less common among older female adults compared with

Table 3. Associations of Orthostatic Hypotension (OH) Phenotypes With Orthostatic Symptoms (N=863*)

OH phenotypes	Odds ratio (95% CI)					
	Symptoms during the OH assessment (concurrent)		Symptoms in the last 30 d (historic)			
	Symptoms in the process of standing	Symptoms at any time during the standing position	Lightheadedness	Dizziness	Fainting/blackout	Imbalance
N, %	87 (10.1)	49 (5.7)	335 (38.8)	292 (33.8)	29 (3.4)	538 (62.3)
Initial OH*						
Yes	2.34 (1.42–3.86)†	1.82 (0.93–3.57)	1.48 (1.08–2.02)†	1.34 (0.97–1.84)	0.94 (0.41–2.16)	1.32 (0.96–1.82)
No	Reference	Reference	Reference	Reference	Reference	Reference
Early OH						
Yes	1.82 (1.11–2.98)†	1.94 (1.02–3.69)†	1.29 (0.92–1.81)	1.30 (0.93–1.84)	1.60 (0.70–3.64)	1.06 (0.75–1.50)
No	Reference	Reference	Reference	Reference	Reference	Reference
Late OH						
Yes	1.29 (0.73–2.27)	1.43 (0.68–3.02)	1.24 (0.85–1.81)	1.26 (0.86–1.84)	1.40 (0.56–3.51)	0.92 (0.62–1.36)
No	Reference	Reference	Reference	Reference	Reference	Reference
Sustained OH						
Yes	1.33 (0.73–2.41)	1.54 (0.71–3.34)	1.35 (0.91–2.01)	1.33 (0.89–1.99)	1.09 (0.39–3.05)	0.96 (0.63–1.45)
No	Reference	Reference	Reference	Reference	Reference	Reference

There are 24 models presented in this table. Each model included an OH phenotype as the independent variable and symptoms as the dependent variable. All models were adjusted for age, body mass index, time to stand, sex, race-study center, diabetes, hypertension and antihypertensive use, hypnotic use, prior coronary heart disease, prior stroke, and frailty using logistic regression. Results were considered statistically significant if the 95% CI did not include the null value.

*n=773 (90 participants missed the first standing measurements).



older male adults. Higher prevalence of OH among older male adults could be due to a higher burden of cardiovascular diseases, as well as differences in comorbidities and medication use.^{40–42} Further research is required to understand the interactions between OH, age, and sex, which may inform targeted screening strategies for early diagnosis of OH.

The association between antihypertensive medications and OH remains inconsistent across the literature.⁴³ Several observational studies have reported strong

positive associations between various antihypertensive classes and OH,^{30,31,43–45} whereas others reported no association^{1,2,46} or even reduced risk.⁴⁷ However, national guidelines caution against initiating antihypertensive therapy, especially in adults with prior falls.^{11,43,48} In our study, antihypertensive use was strongly associated with OH regardless of BP control. This may not be surprising because certain commonly used antihypertensives, such as α - or β -blockers, lower BP by inhibiting sympathetic stimulation to the heart and blood vessels.³⁰ This

Table 4. Associations of Orthostatic Hypotension (OH) With Falls

OH phenotypes	Falls in the last 12 mo (historic), N=843		Falls during follow-up (prospective), N=772	
	Rate (no. of falls per person)	Rate ratio (95% CI)	Rate (no. of falls per person)	Rate ratio (95% CI)
Initial OH*				
Yes	0.40	1.27 (0.94–1.71)	0.46	1.45 (1.02–2.06)†
No	0.30	Reference	0.27	Reference
Early OH				
Yes	0.42	1.17 (0.87–1.58)	0.46	1.08 (0.74–1.60)
No	0.34	Reference	0.36	Reference
Late OH				
Yes	0.38	1.01 (0.71–1.42)	0.47	1.10 (0.72–1.69)
No	0.35	Reference	0.36	Reference
Sustained OH				
Yes	0.39	0.98 (0.68–1.41)	0.46	1.00 (0.64–1.57)
No	0.35	Reference	0.37	Reference

There are 8 models presented in this table. Each model included an OH phenotype as the independent variable and falls as the dependent variable. All models were adjusted for age, body mass index, time to stand, sex, race-study center, diabetes, frailty, prior coronary heart disease, prior stroke, hypertension and antihypertensive use, and hypnotic use using negative binomial. Results were considered statistically significant if the 95% CI did not include the null value.

*n for initial OH and falls in the last 12 mo=755, n for initial OH and falls during follow-up=688.

effect may be intensified at very old ages and unmask OH, because they are more likely to have age-related autonomic impairments and reduced heart function.³¹ However, discontinuing antihypertensive medications may worsen BP and increase the risk of adverse events and orthostatic symptoms.⁴⁹ Further research is needed to clarify whether OH arises from hypertension itself or its treatment.

Our study showed that initial and early OH were associated with orthostatic symptoms during the process of standing, whereas sustained OH was not. Prior work among a middle-aged ARIC population demonstrated similar findings with respect to earlier OH and orthostatic symptoms.⁸ Similarly, among older adults at risk for falls from the STURDY trial (Study to Understand Fall Reduction and Vitamin D in You), early OH was more strongly associated with orthostatic symptoms.¹² Although mechanisms were beyond the present study, we can speculate as to reasons for this observation. OH within 3 minutes of standing (initial and early) may reflect an acute impairment, whereas OH beyond 3 minutes (sustained) may suggest chronic dysregulation. As a result, older adults with sustained OH may have developed tolerance over time, resulting in fewer perceived symptoms.^{50–52} In addition, symptoms depend on the ability to autoregulate blood flow to the brain, which can be impaired in some older adults with chronic autonomic impairment.⁵³ Future research should confirm this hypothesis, however.

Our study showed that older adults with initial OH were more likely to have falls, consistent with previous findings.^{1,8,32} A study by Juraschek et al⁸ showed that OH within 1 minute was strongly associated with orthostatic symptoms and falls. Another study reported that older adults with uncontrolled hypertension and systolic OH at 1 minute had a higher risk of falls within 1 year.¹ However, in the STURDY trial, OH immediately after standing had higher risks of falls, but the association was not significant, whereas OH beyond 3 minutes was more strongly associated with falls. These discrepancies could be due to differences in sample sizes and fall assessment methods.¹² Additional research is needed to replicate our findings.

Our study has several limitations. First, the generalizability of our findings may be limited to less frail older adults, because this study included White and Black adults with an average age of 83.7 years selected from 4 communities in the United States, who were able to complete the OH assessment. Second, we were missing BP measurements immediately after standing for some participants, which might have underestimated initial OH and its association with orthostatic symptoms and falls if the measurements were missing due to OH. Nevertheless, our sensitivity analysis treating these measurements as OH was reassuring. Third, falls in the prior 12 months or orthostatic symptoms during the prior 30 days were self-reported and may be subject

to recall bias, or truly symptomatic participants may have declined to attend the visit due to fall concerns. However, our findings on prospective falls and concurrent orthostatic symptoms are robust. Fourth, precision was poor for some analyses due to very small sample sizes. Fifth, an investigation of individual antihypertensive medication classes with OH, orthostatic symptoms, or falls was beyond the scope of the study and should be the focus of future work. Sixth, orthostatic symptoms were selected based on the validated Orthostatic Hypotension Questionnaire⁵⁴ and the STURDY trial.⁵⁵ However, these populations were not exactly similar to the ARIC; for example, the OHQ was validated among patients with neurogenic OH (mean age, 63 years) and the STURDY trial enrolled free-living adults (mean age, 76 years) at risk for falls. Given the lack of a validated OH symptom questionnaire for the ARIC population, misclassification of symptom status may have occurred. Finally, as an observational study, our findings may be influenced by residual confounding.

Our study has several strengths. First, to our knowledge, this is the first study that conducted comprehensive temporal assessments of OH and orthostatic symptoms among very old adults in the United States. People of this age group are often underrepresented in clinical research and have a high burden of hypertension and BP variability.⁵⁶ Second, this study was conducted by rigorously trained staff using standardized protocols of the OH assessment and structured questionnaires, ensuring high-quality data collection. Third, ARIC's comprehensive and robust data collection allowed us to adjust for relevant clinical covariates, which minimized confounding.

In conclusion, in this community-based cohort population of very old ages, we found that OH within 3 minutes of standing, including initial and early OH, was significantly associated with orthostatic symptoms and falls; moreover, elderly male and those treated for hypertension had higher risks of OH. These findings emphasize that clinicians should measure BP within 3 minutes of standing, particularly immediately after standing, to identify OH among people at very old ages. More intensive monitoring for OH, including BP measurements immediately after standing, is particularly important for men and for those receiving antihypertensive medications. Future research may explore the associations between specific classes of antihypertensive medications and OH to better inform treatment strategies.

PERSPECTIVES

In this community-dwelling population of adults at very old ages, OH within 3 minutes of standing, such as initial and early OH, was common. These findings underscore the importance of measuring BP immediately after standing, which is sometimes dropped from

traditional OH assessments. This study also highlights the complexity of hypertension management and the risk of developing OH and orthostatic symptoms among older adults. Future research should examine whether OH found among older adults is primarily driven by antihypertensive treatment or underlying hypertension. In addition, studies should investigate whether specific classes of antihypertensive medications are more strongly associated with OH, to inform tailored strategies for reducing OH risk during hypertension management among older adults.

ARTICLE INFORMATION

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Disclosures

None.

Supplemental Material

Supplemental Methods

Tables S1–S17

Figures S1–S4

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